Introduction

Advanced Machine Learning

Sam Ogden

Class Overview

What is Advance Machine Learning?

- This course is meant to take us from linear regression models up to the concepts of Large Language Models
  - And a number of other places, too!

Course Learning Outcomes

- Be able to understand what machine learning is, reason about it, and apply it
- Define key algorithms, structures and techniques such as:
  - `SGD`
  - `Ensemble Methods`
  - `Dimensionality Reduction`
  - `MLPs`
  - `CNNs`
  - `LLMs`
- Use `PyTorch` to build, train, and deploy deep learning models

Teaching Staff

Me (Dr. Sam Ogden)

- BS (Math&EE), Univ. of Vermont
- MS (CS), Univ. of Vermont
- PhD (Comp. Sci.), WPI

- Product Engineer @ IBM/GF (5 years)
- Grad Student @ WPI (5 years)
- Professor @ CSUMB (since 2022)

- Rock climbing
- Running
- Reading
- Getting lost (“exploring”)
- Ice cream
- Good bread
- Nerdy questions

- Office hours:
- Tuesdays 10-11am
- Wednesdays 1-2pm
- By Appointment

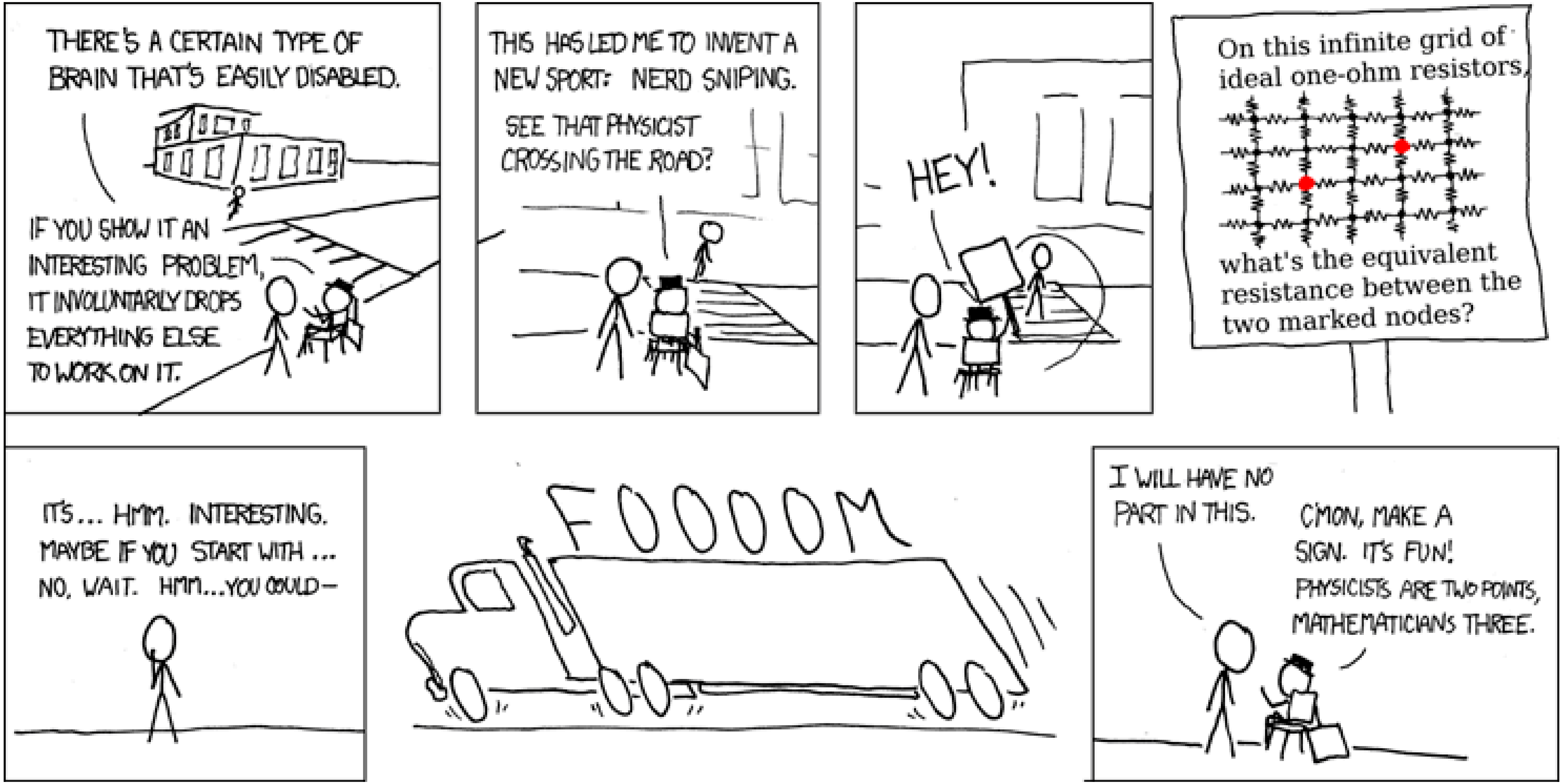
Asking Questions

I love getting asked questions

During class please raise your hand and I’ll get to you as soon as I can

If you’re confused then chances are that you’re not the only one

Questions: Nerd Sniping

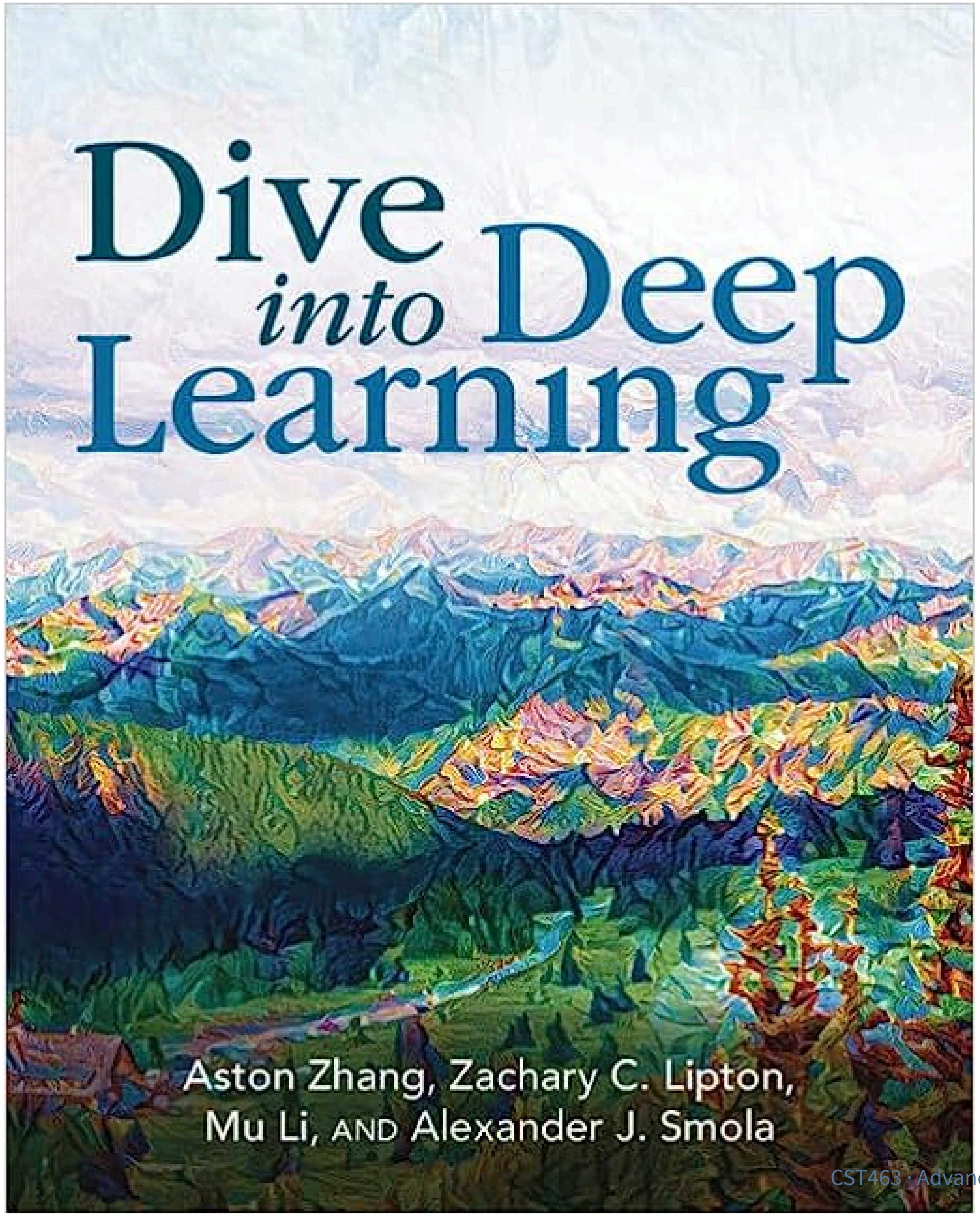


Course TAs

Cat Tax

Course Information

Course Textbook: Dive into Deep Learning



Available online for free

- Also as a PDF
- We'll be using the PyTorch version of the book

It's goal is to walk from Linear Regression to Modern large-scale Deep Learning models

Lots of code examples, practice problems and material available for it

Course Schedule (might change a bit!)

Class Structure

The goal is for most days to have a lecture, a break and a lab:

- lecture (50 minutes)
- break (10 minutes)
- lecture (35 minutes)
- lab (15 minutes)

(but if you've ever had a class with me you know this is only a goal)

Course Policies

Link to the Syllabus

Grading

Assignment Kind	Value
Homework	36%
Exams	36%
Final Project	16%
Participation	12%

You must get a 60% overall, and 40% in each subset of these in order to pass

Attendance

Attendance is expected and enforced through quizzes

Use of LLMs (1)

My personal thoughts on LLMs can be summarized as:

- LLMs are very useful tools for accelerating our work.
- LLMs are fantastic at making “interactive data”
- LLMs can be incredibly stupid. (And see the first bullet…)

One of the things are is most valuable about us is **our ability to problem solve**.

But we have to **learn how** and **keep practicing** it

Use of LLMs (2)

How I expect you to use LLMs in this class:

- Debug computer problems
- Better understand the material
- Quick iteration
- Checking your approach and assumptions

How I expect you to *not* use LLMs in this class:

- Abdicating decisions and/or thinking
- Taking the class for you
- Copy-and-paste

The point of you being in this class is to understand the topic, right?

What gets you a job is the ability to think and put pieces together.

If I ask you to explain your code or analysis, you should be able to do so.

Assignment Details

Homework (30%)

Phase 1

Coding

Write code to understand functionality

- Automatically graded via unit tests

Phase 2

Exploration & Analysis

Use the code to explore behavior

Write a report on parameters and findings

- Graded by peer evaluation based on a rubric

Phase 3

Peer Evaluation

Evaluate your peers's work to help both of you improve communication in reports

Each phase will be 1 week, but phases 1 & 3 will be overlapped

Exams (30%)

- 2-3 equally weighted exams
- closed book, but 1 handwritten page of notes
- Material builds, so effectively culumulative
- Practice Quizzes will likely be provided beforehand

Final Project (20%)

The goal is to play around with a topic you find interesting that's within the scope of the class

- Research Project (10%):
  - Group project that will have checkpoints through the semester

Participation (20%)

- Class Attendance (10%):
  - many classes start with attendance quizzes
  - reading is linked on weekly schedule
  - More likely to have quizzes when class looks empty
- Weekly Study Notes (5%)
  - Intended to help you write yourself a study guide for exams
  - Reflect on the topics covered this week
    - what you learned this week
    - what you struggled with
  - Graded largely on completion
  - The guidelines are meant as an algorithm – you can use whatever format you find most helpful to study from
- Lab activities (5%):
  - due Sunday night after in-class introduction

Weekly Study Notes: help your future self

These study notes are for you to go back and use as a basis for studying

How to approach these notes:

1. identify the topics that were covered
2. explain ideas in your own words
3. identifying least-understood topics
4. noting “aha” moments
5. explaining the nature of the confusion
6. asking questions
7. wondering about ideas, connections, applications
8. deeper reflection

Homework

How to get help

TA Office Hours!

How to get help: what's okay

- Asking your instructor!
  - Seriously, come by office hours
- Asking your TAs!
  - They've taken the class and are nice!
- Asking LLMs for clarification or guidance!
  - "What does limited direct execution mean?"
  - "Can you make me some practice problems for process scheduling?"

But don't trust LLMs entirely!

How to get help: what's not okay

- Getting `[fill in the blank]` to write code for you
  - Friends
  - Enemies (for numerous reasons this is a bad idea)
  - LLMs
- The goal is to understand the problems and learn how to approach them!
  - Any code you turn in you are responsible for!

Dealing with Challenges in Class

Take a breath

Mental Health

- Mental health is super important
- If you're feeling stressed or anxious about an assignment then take a break on it for a bit!
  - (This is the real reason I want you to look at assignments early — so you have time to walk away for a bit)
- If you're feeling stuck then ask me or the TAs!

How to solve hard problems

Give yourself time to solve problems

College has a lot going on, but starting early gives you time to step back

Sometimes we run into impossible problems and need to start over

Things to do

- Go for a walk without devices
- Go to the gym
- Stare at a wall
    - (boredom is your friend)
- Read a dumb book

Things to not do

- Screens.
    - (I really could stop there)
- Use your brain to focus heavily

Give yourself time and space to let your mind wander and make random connections

A quick note back to LLMs

This is one of the big traps with LLMs – they force you into *doing* rather than *thinking*.

- I’ve seen this myself and know a lot of pros are running into it, too

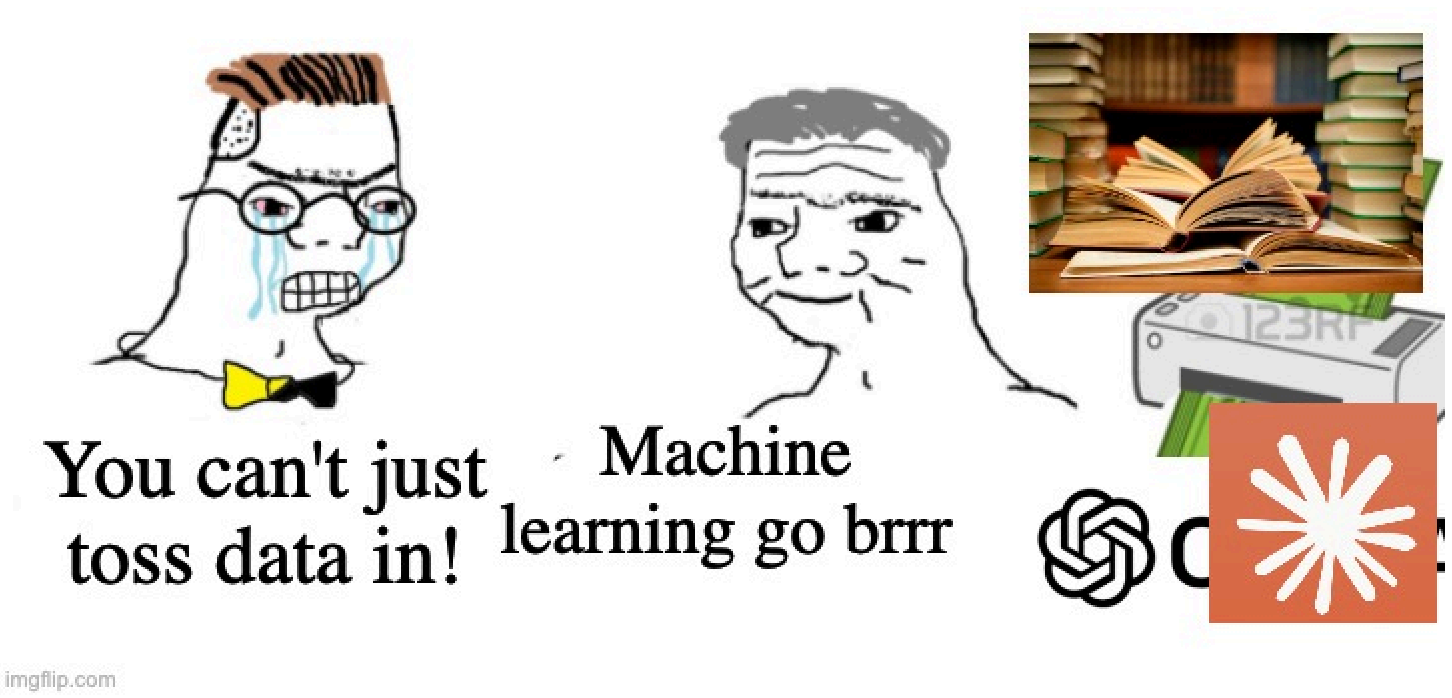
This causes burnout and poor learning

Some things to remember

- Focus on growth and mastery
- Start homework early
- Drop by my office to chat
- Please call me just Sam (or Dr./Prof. Ogden)

Advanced Machine Learning

What is Machine Learning



How to think about machine learning

Machine Learning works on feeding **data** into a **model** that we think fits the data

Critical things:

- We propose a *model shape*
  - This is our hypothesis
- The data is a *constant*
  - We use it to determine the best model parameters
- We want to measure how well we are fitting our parameters

Modeling Data

Things we need to think about

- What model shape makes sense for our data?

- How do we quantify how well we’re doing?

- Do we need as much complexity as we have?

- How can we run this reasonably?

Welcome to Advanced Machine Learning!

Lab time!